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The role of self-efficacy and curiosity in student use of artificial intelligence (AI)

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Abstract
Availability and use of artificial intelligence (AI) is rapidly increasing, presenting both challenges and opportunities for students. As AI provides opportunities and benefits for students, it is useful to examine what factors are associated with student use of AI. Two of these factors are self-efficacy and curiosity. This study investigated the connections between self-efficacy for use of AI, curiosity, and reported use of AI among university students. One hundred and twenty three university students participated in the study. Their mean age was 37.83, *SD* = 20.64, and 88 were women. Participants reported the amount of time they spent using AI in their university studies and completed measures of self-efficacy for use of AI and curiosity. Self-efficacy for use of AI was associated with more use of AI. A higher level of general curiosity was associated with greater self-efficacy for use of AI, with the exploration dimension of curiosity showing an especially strong association with self-efficacy. Self-efficacy for use of AI linked the association of curiosity with use of AI. These findings regarding self-efficacy and curiosity provide new information relevant to the use of emerging AI technology. Insight into links between self-efficacy and curiosity and use of AI can assist educators in facilitating the benefits of student AI use.

Keywords Artificial intelligence, Broaden and Build theory, Curiosity, Higher education, Self-efficacy, Student technology use, Theory of planned behavior

Introduction

A technology available to many students is artificial intelligence (AI). Examples of AI programs useful to students include ones such as ChatGTP and Gemini for information searches, Grammarly for advice regarding improving spelling and writing style, and Kickresume for guidance in building resumes (Adwani, 2025). Giannakos et al. (2024) and Bozkurt and Gursoy (2025) pointed out that use of AI creates opportunities as well as challenges. For students, opportunities can include self-regulated learning and collaborative learning (Giannakos et al., 2024). As mentioned by Wang et al. (2024a), AI enhances accessibility of information, personalization of information, time saving through automation, and interactivity in the learning process. Thus, when used effectively, AI benefits students. Accordingly, it is important to examine what student characteristics influence students' use of AI. Self-efficacy and curiosity may underpin such use.

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Literature review

As use of AI benefits students, it is useful to examine what factors are associated with student use of AI. Some research has investigated student characteristics associated with use of AI. For example, Rajki et al. (2025) found that male students reported use of a wider variety of AI tools and experimented more with AI technology, while women reported more barriers related to AI use. Lien et al. (2025) and Wang et al. (2024b) found that more positive attitudes towards AI were linked to more use of AI and intention to use AI. Acosta-Enriquez et al. (2024) found that perceived ease of use of AI technology and acceptance of AI tools were significant predictors of intentions to use AI. Alshammari et al (2025) reported that students' perception that AI would enhance their performance was related to their intentions to use AI.

Two important psychological characteristics, self-efficacy and curiosity, have been investigated in relation to AI. Asio (2025) found that students' positive attitudes towards AI, understanding of AI tools, and self-efficacy for using AI were inter-related. Bewersdorff et al. (2025) reported that interest in AI together with understanding of AI enhanced AI -related self-efficacy. In a study that included students from 47 countries, Gillespie et al. (2025) found that curiosity regarding new technologies predicted exploration and use of AI. Based on a survy of over 1,000 students, Freeman (2025) reported that curiosity leads to investigation of AI and use of AI. As self-efficacy and curiosity are linked to student engagement with AI, more information regarding the roles of self-efficacy and curiosity in relation to use of AI will expand understanding of factors influencing adoption of AI tools.

Self-efficacy

Self-efficacy is the belief that one can bring about desired outcomes (Bandura, 1986, 2012). Self-efficacy can be general or realm-specific, and both aspects of self-efficacy promote optimal functioning in various areas of life (Schutte & Malouff, 2016). General self-efficacy is the belief that one can bring about good outcomes in various areas of life and deal with a variety of opportunities and challenges (Luszczynska et al., 2005). Realm specific self-efficacy is the expectation of being able to achieve desired outcomes in a specific realm of life, such as sleep (Ghose et al., 2023) or in relation to role-specific tasks, such as a teacher's self-efficacy for literacy instruction (Schwab et al., 2023).

Self-efficacy is a predictor of students' success. In a large meta-analysis of 231 studies that included over 242,000 participants, Çikrıkci (2017) found that across studies, a higher level of self-efficacy was related to greater student success. The relationship between self-efficacy and success in this realm may be bi-directional, as suggested by a meta-analysis of student performance (Talsma et al., 2018). In this meta-analysis that included over 2,000 participants across studies, Talsma et al. found reciprocal relationships between self-efficacy and academic performance in two wave longitudinal studies. Performance predicted later self-efficacy and self-efficacy predicted later performance.

An aspect of student self-efficacy is confidence regarding use of technology. For example, Bates and Khasawneh (2007) found that university students' online learning self-efficacy was associated with time spent using on-line learning technologies to complete assignments. Getenet et al. (2024) found that university students with higher levels of self-efficacy for use of digital technology showed greater learning engagement, including behavioral, social, collaborative, and cognitive engagement.

Curiosity

Curiosity is a positive emotion with motivating characteristic (Tang et al., 2022) that prompts the desire to know and learn (Silvia, 2012). Curiosity consists of related dimensions (Kashdan et al., 2018), including desiring exploration for its own sake (Kashdan et al., 2018; Litman, 2005), deprivation sensitivity or wanting to decrease gaps in knowledge (Kashdan et al., 2018; Litman, 2005), and tolerance of stress when faced with novelty (Kashdan et al., 2018). Curiosity is a foundation for student success (Markey & Loewenstein, 2014). The link between greater curiosity and better student performance has been found in early childhood cohorts (Shah et al., 2018), among secondary students (Tang & Salmela-Aro, 2021) and university students (Shin, 2024).

Shin (2024) found an interaction between curiosity and self-efficacy in predicting academic outcomes among university students. Both higher self-efficacy and higher curiosity assessed at the start of a semester predicted better later performance as assessed through student grades. Among students with low self-efficacy, a higher level of curiosity buffered the effect of this lower self-efficacy on grades. Curiosity is a foundation for self-efficacy in various realms. For example, Chang et al. (2025) found that curiosity, especially the exploration-related dimension of curiosity, predicted entrepreneurial self-efficacy, and this self-efficacy in turn predicted outcomes such as starting businesses.

Arnone et al. (2011) proposed that curiosity is an important foundation for student engagement with technology. This engagement with technology stimulated by curiosity enhances inquiry-based learning and deep learning (Arnone et al., 2011). Curiosity is one of the most predominate reactions to technology represented by AI (Sago, 2025).

The Broaden and Build theory

The Broaden and Build theory (Fredrickson, 2013) offers a conceptual framework for understanding possible ties between curiosity, self-efficacy, and use of AI among students. The Broaden and Build theory posits that positive emotions such as joy, interest, and curiosity expand and broaden individuals' awareness and divergent thinking. This broadening prompts exploration of thoughts and new environments. A result of this broadening is the building of competencies and beneficial characteristics. Results of experimental research support the broadening impact of positive emotions leading to beneficial outcomes such as greater resilience and increased mindfulness, which in turn are linked to decreases in depression and increases in life satisfaction (Fredrickson et al., 2008). Increased self-efficacy is one of the characteristics impacted by broadening through positive emotions, and such increases in self-efficacy are in turn associated with better mental health and greater relationship satisfaction (Schutte, 2014). Supporting the proposition that broadening through positive affect may impact use of AI, Acosta-Enriquez et al. (2024) found that general experience of positive emotions was linked to intentions to use AI.

Previous research and theory associated with self-efficacy and curiosity suggest that these characteristics are relevant to students' use of AI. Further research findings may offer information regarding how they individually and in interaction relate to use of AI. The present research set out to provide this information.

Aim of the present study

The aim of the present study was to examine the connections between curiosity, self-efficacy for use of AI, and use of AI among university students. The hypotheses underpinning the research were as follows:

1. Greater self-efficacy for use of AI is associated with more use of AI.
2. A higher level of general curiosity is associated with more use of AI.
3. A higher level of general curiosity is associated with greater self-efficacy for use of AI.
4. Self-efficacy for use of AI connects general curiosity with use of AI.

Methodology

Participants and procedure

The study was approved by the Human Research Ethics Committee of the University of New England, Australia. One hundred and twenty three Australian university students participated in the study. They were recruited through announcements on university web pages. Their mean age was 37.83, *SD* = 20.64, with a wide range of ages from 17 to 70. The sample included 88 women (72%), 33 men, and two who did not indicate gender. Participation was online, with participants providing information through the Qualtrics research platform. One hundred and forty six students originally considered participation, as indicated by their accessing the on-line participant information sheet and consent form, but seven of these potential participants elected not to progress to the study and 16 discontinued participation after entering the study. Participants could only progress through the study after responding to each survey item.

Materials

Participants reported how many hours they spent per week usingAI programs as part of their university work and completed measures of curiosity and self-efficacy for use of AI.

Time spent using AI for university work

After being provided with some examples of common AI programs, such as ChatGTP and Grammarly, participants recorded how many hours they spent in a typical week using AI programs in their university work. The mean number of hours reported was .70, *SD* = 1.65. The range of hours was substantial at 0–15 h.

Curiosity

General curiosity was assessed through the exploration and deprivation scales of the The Five-Dimensional Curiosity scale (Kashdan et al., 2018). These dimensions account for the most variance in curiosity, and the scale and subscales have previous evidence of reliability and validity (Kashdan et al., 2018). An example of an item assessing exploration related curiosity is "I enjoy exploring new ideas and experiences." An example of an item assessing deprivation related curiosity is "I feel uncomfortable when I don't know something."

In the present study, internal consistency was alpha = .81 for overall curiosity assessed through the combined 14 items that report exploration and deprivation related curiosity, alpha = .91 for the seven items assessing exploration related curiosity, and alpha = .59 for the seven items assessing deprivation related curiosity. Thus, reliability for

overall curiosity and exploration related curiosity was good, but reliability of assessment of deprivation related curiosity was poor.

Self-efficacy for use of AI

An eight-item scale, based on Bandura's (2006) guide for construction of realm-specific self-efficacy scales, was created for the study to assess self-efficacy for use of AI. Participants rated their confidence in use of various aspects of AI. Sample items include 'Find the right AI program for what I need at the time', 'Create a follow-up prompt when an AI program provides information and I need further information', and 'Verify information provided by an AI program.' Internal consistency for the scale was alpha = .95, indicating good reliability.

Results

Analysis plan

Pearson correlation analyses, with two-tailed significance levels and 2000 bootstraps to take into account normality of data assumptions, tested the hypotheses that

1. Greater self-efficacy for use of AI would be associated with more use of AI,
2. A higher level of general curiosity would be associated with more use of AI, and that
3. A higher level of general curiosity would be associated with greater self-efficacy for use of AI.

A mediation analysis using the PROCESS program model 4 (Hayes, 2022), with 2000 bootstraps, tested the hypothesis that

4. Self-efficacy for use of AI would connect general curiosity with use of AI.

Post hoc analyses investigated whether the curiosity dimensions of exploration and deprivation would show differential connections to self-efficacy and use of AI.

Results for testing of hypotheses 1–3

A Pearson correlation testing the hypothesis that greater self-efficacy for use of AI would be associated with more use of AI found $r(122) = .27$, $p = .01$. This indicated that greater self-efficacy for use of AI was significantly associated with more time spent using AI as part of university work.

A correlation testing the hypothesis that a higher level of general curiosity would be associated with more use of AI found $r(122) = .05$, $p = .62$, indicating that general curiosity was not significantly associated with AI use. Post hoc analyses investigated the connections of the dimensions of exploration related curiosity and deprivation related curiosity with use of AI. Results showed neither to be significantly associated with use of AI; however, exploration curiosity, at $r(122) = .13$, had a higher association than deprivation curiosity at $r(122) = .06$.

A correlation testing the hypothesis that a higher level of curiosity would be associated with greater self-efficacy for use of AI found $r(122) = .21$, $p = .03$, indicating that more curiosity was significantly associated with greater self efficacy. Post hoc analyses examined the ties of the exploration and deprivation dimensions of curiosity with self-efficacy. Exploration related curiosity, at $r(122) = .28$ and $p = .01$, showed a significant

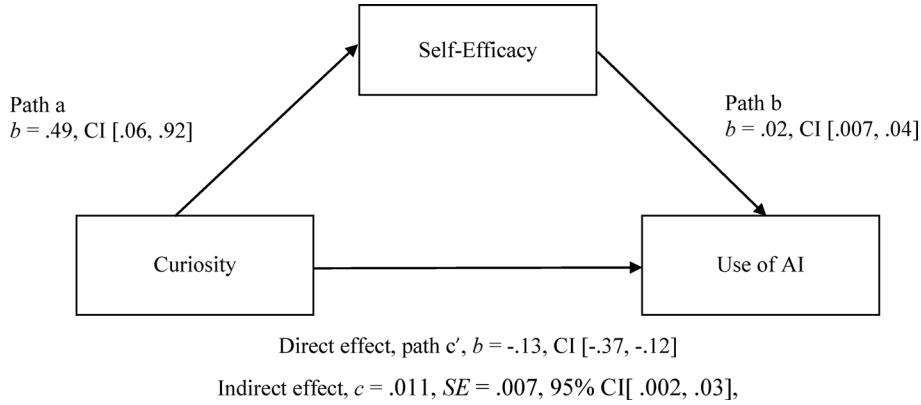

Fig. 1 Model and mediation results. *Note*. Based on 2000 bootstrap resamples. Values are unstandardised beta coefficients. CI is 95% confidence interval

association, while deprivation related curiosity, at $r(122) = .06$, did not show a significant association.

Results for testing of hypothesis 4

A mediation analysis tested the hypothesis that self-efficacy for use of AI would connect general curiosity with use of AI. The overall model was significant, $R^2 = .08$, $F(2, 118) = 4.72$, $p = .02$. The indirect effect of the mediation model was significant, with an effect of .011, $SE = .007$, 95% CI [.002, .03], indicating that self-efficacy linked curiosity with use of AI (see Fig. 1). As exploration related curiosity showed the strongest associations with use of AI and self-efficacy, a post hoc analysis investigated the self-efficacy link between this aspect of curiosity with use of AI. The indirect effect of this mediation model was significant, with an effect of .02, $SE = .01$, 95% CI [.006, .05], indicating that exploration related curiosity was an especially important predictor in the self-efficacy link between curiosity and use of AI.

Discussion

The present study investigated factors associated with use of artificial intelligence (AI) among university students. Self-efficacy for use of AI was associated with more use of AI. A higher level of general curiosity was associated with greater self-efficacy for use of AI, with the exploration dimension of curiosity showing an especially strong association with self-efficacy. In regard to the connections between curiosity and self-efficacy in their relationship to use of AI, self-efficacy for use of AI linked curiosity, and especially the exploration dimension of curiosity for use of AI. The model examining the connections between curiosity, self-efficacy and use of AI was significant. However, the R^2 value for the model indicated that only a portion of the variance in AI use was explained by the model. A number of other factors and student characteristics likely contribute to student use of AI.

The results of the present study can be interpreted through the framework of the Broaden and Build theory. Greater curiosity can lead to broadening, prompting development of self-efficacy in use of AI. This self-efficacy may encourage engagement with AI, as represented by more time spent using AI. The Theory of Planned Behavior (Ajzen, 1985) also provides a framework for interpreting the results of the study. The theory posits that perceived behavioral control, which is similar to the construct of self-efficacy,

is a key determinant of intention to perform a behavior, and then intention in turn is a strong predictor of actual behavior. Taken together, the finding of the present study that self-efficacy is associated with AI use and the finding of Wang et al. (2024b) that demonstrates the role of intention in AI use align with the Theory of Planned Behavior. Accordingly, this interpretation suggests examining additional factors, such as normative beliefs and evaluation of outcomes, that the theory proposes to underlie behavior.

The findings of the present study together with theoretical contexts provided by the Broaden and Build theory and the Theory of Planned Behavior have pedagogical implications. If educators wish to encourage students' beneficial use of AI, they might consider strategies to prompt curiosity in regard to AI and to encourage self-efficacy in use of AI through providing opportunities for mastery experiences when using AI tools and providing examples of successful AI use. Building student self-efficacy, along with other factors proposed by the Theory of Planned Behavior to underlie behavioral intentions, can then result in student's beneficial use of AI.

Limitations and future research

The present study used a concurrent data collection methodology; thus, even though the hypotheses tested have underlying assumptions regarding causal connections between curiosity, self-efficacy, and use of AI, no definite causal conclusions can be drawn. Further, curiosity, self-efficacy and use of AI were assessed through self-report. Using the same method and source, such as concurrent self-reports taken in this study, to measure an association between variables, can lead to a bias. According to Podsakoff et al. (2012), common method and source bias can lead to stronger assessed associations than the actual associations.

In regard to measurement of constructs, the curiosity-deprivation dimension showed low reliability, and overall AI usage was low and highly dispersed, which together possibly limit the validity of assessment of AI use. Finally, participants in the study were students from one country. The experiences of students from other countries or cultures and participant samples with different demographic characteristics, such as different gender composition or ages, may vary from those of the study sample.

Future research might explore the impact of increases in curiosity on development of self-efficacy for use of AI. Such research could involve longitudinal data collection that records the impact in changes in curiosity, self-efficacy, and actual use of AI over time. Future experimental research can manipulate curiosity (general, topic-specific, or AI-specific) to assess the impact of increased curiosity on self-efficacy for use of AI, intention to use AI and actual use of AI. Additional research can investigate the interplay between self-efficacy, curiosity, and other characteristics, such as achievement motivation, optimism, and factors identified in the Theory of Planned Behavior, in relation to AI use. Future research can also examine the relationships between curiosity, self-efficacy, and AI among students in different countries.

Conclusion

In conclusion, self-efficacy for use of AI is linked to use of this emerging technology. Curiosity is a motivating emotion that underlies the connection of self efficacy with use of AI. As availability of AI programs is growing and use of AI is rapidly increasing (Bozkurt & Gursoy, 2025), presenting both challenges and opportunities for students, it is

important to understand factors associated with student use of this technology. Insight into characteristics linked to use of AI, such as self-efficacy and curiosity, can assist in facilitating the beneficial use of AI.

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Author contributions
NS and HL jointly conceptualized the study, analyzed data, wrote the manuscript describing the study, and edited the manuscript.

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Data availability
Data is available from the corresponding author upon request.

Declarations

Competing interests
The authors declare no competing interests.

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